

LAMPANG AGROMET, Thailand

WMO: 483340

Lat: 18.3250N

Lon: 99.3014E

Elev: 319

StdP: 97.55

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 13.1	(c) 14.8	(d) 9.3	(e) 7.6	(f) 21.9	(g) 10.5	(h) 8.2	(i) 21.5	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 12.6	(c) 38.6	(d) 23.9	(e) 37.1	(f) 23.8	(g) 35.9	(h) 23.6	(i) 27.1	(j) 31.8	(k) 26.7	(l) 31.3	(m) 26.5	(n) 31.0	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.1	22.3	29.0	25.7	21.8	28.6	25.5	21.5	28.4	88.0	32.2	86.5	31.5	85.3	31.0	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	22.4	24.8	27.8	30.0	29.0	28.2	27.7	27.3	27.3	26.5	25.3	22.5
	DBStd	2.86	2.09	1.83	2.18	2.22	1.86	1.25	1.43	0.93	1.01	1.16	1.41	2.36
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	7	3	0	1	0	0	0	0	0	0	0	0	3
	CDD10.0	6046	383	413	551	599	589	547	550	536	520	511	459	389
	CDD18.3	3011	128	180	292	349	331	297	292	278	270	252	209	134
	CDH23.3	31066	1027	1928	3582	4586	3744	3188	2964	2588	2573	2184	1726	973
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	PrecAvg	1116	10	9	28	65	172	122	146	203	212	106	32	11
	PrecMax	1691	71	47	117	171	305	203	241	358	399	183	154	96
	PrecMin	778	0	0	0	9	25	37	74	85	103	30	0	0
	PrecStd	195	18	13	30	37	78	42	47	67	61	41	33	20
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	35.5	38.3	41.0	38.8	35.7	36.2	33.4	33.6	32.9	32.5	31.5
		MCWB	21.2	20.6	22.1	23.6	25.4	26.2	25.5	25.9	26.6	25.8	25.0	23.5
	2%	DB	30.5	34.2	37.1	39.7	36.8	34.2	34.1	32.3	32.6	32.0	31.6	30.3
		MCWB	21.2	21.0	22.1	23.9	25.4	26.1	25.6	25.9	26.4	25.7	24.5	23.0
	5%	DB	29.5	33.1	36.1	38.2	35.2	33.1	32.7	31.5	31.8	31.3	30.8	29.3
		MCWB	21.2	21.3	21.8	23.9	25.6	26.0	25.7	25.9	26.2	25.4	24.2	22.6
	10%	DB	28.5	31.9	34.9	36.7	33.8	32.0	31.5	30.6	30.9	30.5	30.0	28.2
		MCWB	20.9	21.2	21.9	24.2	25.7	25.9	25.7	25.8	26.0	25.3	24.1	22.1
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.4	24.2	25.1	26.5	27.5	27.3	27.2	27.1	27.5	26.7	26.0	25.2
		MCDB	28.4	30.9	33.4	35.0	33.3	32.3	31.2	30.9	31.7	30.8	30.5	29.0
	2%	WB	23.3	23.6	24.3	25.7	26.8	26.9	26.7	26.7	27.0	26.4	25.3	24.4
		MCDB	27.5	30.4	32.4	34.5	32.6	31.8	30.8	30.3	31.3	30.3	29.6	28.3
	5%	WB	22.6	23.0	23.8	25.2	26.4	26.5	26.3	26.4	26.6	26.0	25.0	23.5
		MCDB	26.9	29.8	31.7	33.8	32.1	31.1	30.4	29.9	30.6	29.8	29.2	27.5
	10%	WB	21.8	22.2	23.2	24.8	26.1	26.2	26.0	26.1	26.2	25.6	24.7	22.8
		MCDB	26.3	28.7	31.3	33.1	31.7	30.4	30.1	29.5	29.9	29.2	28.5	26.8
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	15.2	14.3	12.6	9.2	7.4	7.1	6.8	7.3	7.8	9.7	11.2
		MCDBR	14.0	16.7	16.1	14.8	11.5	9.3	9.4	8.3	8.7	9.2	10.4	11.9
		MCWBR	5.7	5.7	4.9	3.7	3.0	2.7	2.7	2.9	3.4	3.7	4.1	5.3
	5% WB	MCDBR	10.7	13.8	13.5	12.4	9.7	8.0	7.7	7.2	8.2	8.1	9.1	10.3
		MCWBR	5.0	5.5	4.5	3.8	3.3	2.8	3.0	3.1	3.6	3.5	3.8	5.0
(40) Clear-Sky Solar Irradiance	taub		0.436	0.514	0.651	0.642	0.493	0.434	0.423	0.437	0.470	0.515	0.439	0.434
	taud		2.201	1.996	1.682	1.763	2.202	2.398	2.427	2.371	2.259	2.088	2.268	2.232
	Ebn at Noon		840	791	699	711	815	856	867	862	834	781	832	829
	Edh at Noon		137	176	249	231	148	120	117	124	138	159	127	129
(44) All-Sky Solar Radiation	RadAvg		4.58	5.31	5.61	6.03	5.44	4.78	4.47	4.41	4.60	4.49	4.61	4.33
	RadStd		0.23	0.20	0.36	0.34	0.47	0.36	0.33	0.28	0.21	0.26	0.31	0.24

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	+0.27	N/A	N/A	N/A	N/A	N/A	+0.15	N/A	N/A	+125	+126

Nomenclature: See separate page